

Summary

1. Which features influence house price the most?

After analyzing the dataset and correlation heatmap, it was observed that **area, number of bedrooms, bathrooms, and stories** have the strongest impact on house prices. Houses with larger areas and more facilities usually have higher prices. Parking availability and furnishing status also affect the price. These features are important because buyers often prefer comfort and space. Therefore, area and facilities are the most influential factors.

2. How accurate was your model (in plain terms)?

The model accuracy was measured using **MAE, RMSE, and R^2 score**. Linear Regression gave a decent prediction, but Random Forest performed better. The Random Forest model was more accurate because it handled complex relationships between features better. A higher R^2 score showed better performance. This means the model can predict house prices fairly well based on the available data.

3. What surprised you in the data?

One surprising observation was that some houses with smaller areas had very high prices. This shows that price does not depend only on size but also on location, furnishing, and extra facilities. Another surprising factor was that furnished houses often had much higher prices compared to unfurnished ones. This proves that many hidden factors can influence the final price.

4. One recommendation for a real estate business based on your findings

Based on the findings, real estate businesses should focus more on properties with larger areas and better facilities because these features increase house value the most. Improving furnishing and parking facilities can also increase selling prices. Builders should analyze customer needs and market trends before pricing properties. Using machine learning can help businesses predict prices more accurately and make better investment decisions.